Owners

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Purpose

The purpose of this BrainLift is to understand how students best prepare for the Physics GRE, in order to build an app that optimizes for the specific skills the exam demands — primarily speed, rapid recall, heuristic reasoning, and interleaved practice.

In Scope

• How the Physics GRE is structured, scored, and administered

• What topics and content appear on the exam and at what emphasis

• How students can build speed and time management skills

• Problem-solving heuristics that work under no-calculator, time-pressured conditions

• The memorization vs. derivation debate and how to resolve it for an app

• What study behaviors and practice patterns produce the best results

• Which sources accurately reflect the post-2023 format

Out of Scope

• Teaching physics concepts from scratch

• Building a textbook replacement

DOK 4: Spiky Points of View (SPOVs)

• **SPOV 1:** The app should prioritize speed over content depth. Several sources already exist to learn physics concepts

• **Elaboration:** Multiple sources converge on the idea that speed is MORE important than content on the PGRE. The exam's brutal time crunch (~1:43/question) and the no-calculator constraint all suggest that the bottleneck is not knowledge but execution speed. An app that drills rapid recall, heuristic application, and question-triage judgment will fill a gap that textbooks and existing resources are unable to address. The insight from "common sense physics" and heuristics (dimensional analysis, limiting cases, numeric substitution) reinforces that many questions don't require full solving at all — they require fast pattern recognition.

• **SPOV 2:** The app should give priority on mastering easier, fundamental concepts rather than teaching difficult, upper level concepts

• **Elaboration:** In several study guides and advice, it is advised to begin with lower level physics material (such as AP Physics C exams) and master those topics before reviewing upper level courses such as nuclear physics or quantum physics. Furthermore, the test is disproportionately weighted towards concepts closer to high school/freshman year physics, such as Classical Mechanics (20% of exam) and Electromagnetism (18%). Also, since getting a “perfect” score only requires around 90% accuracy, a student can afford to fall behind in one of the other subjects, but not in the core subjects. Only exception to this is Quantum Mechanics (12%)

Experts

- **Yoni Kahn and Adam Anderson**
  - **Who:** Physics professors and authors of *Conquering the Physics GRE, Third Edition*
  - **Focus:** GRE-specific tips and tricks; application guide for how to solve problems given existing knowledge; heuristics like dimensional analysis and limiting cases; selective memorization of core formulas
  - **Why Follow:** Authors of the most well-known Physics GRE prep book on the market. Their framework of “derive, don’t memorize” and their heuristic toolkit directly informs the app’s problem-solving skill design
  - **Where:** *Conquering the Physics GRE, Third Edition* (Cambridge University Press, 2018)
- **Lowry A. Kirkby**
  - **Who:** Author of *Physics: A Student Companion*
  - **Focus:** Concept-building approach to physics; the generation effect (equations explained then shown, so students derive before seeing); concise topic overviews with boxed equations
  - **Why Follow:** Their pedagogical approach — derive before reveal — directly informs how the app’s learning sections should be structured
  - **Where:** *Physics: A Student Companion* (textbook)
- **Dr. Jermey Dodd / American Physical Society**
  - **Who:** Presenter of “Acing the Physics GRE: Tips and Strategies” (YouTube)
  - **Focus:** Test-taking strategy; interleaving; making your own formula sheets; common sense physics; testing under real conditions
  - **Why Follow:** Provides practitioner-level strategy advice including the importance of interleaving and the “common sense” elimination approach
  - **Where:** https://www.youtube.com/watch?v=BInkNcJhaWM&t=640s

DOK 3: Insights

- Heuristics (numerically estimating, common sense, question judging) must be prioritized over content breadth. Getting a 990 does not require knowing everything
- The app must weight its content heavily toward lower-division material — textbooks are already accessible, but flashcards beat them for instant recall
- Actual practice should involve as much interleaving as possible. Studying should be done condensed, but doing practice problems should be heavily interleaved

- The app’s value proposition is speed, not content: several sources already exist to learn physics concepts
- Interleaved formulas in drilling mode are non-negotiable, because the exam itself is interleaved
- The learning sections must follow the generation effect: derive first, then explicitly state the formula
- Volume of problems solved is the most reliable predictor of performance — the app must make doing problem after problem the default loop
- Practice tests must be taken under real conditions (early morning, timed, no breaks) to actually simulate the exam, and the app should encourage this
- Reviewing mistakes is non-negotiable: after completing practice tests and problems, thoroughly reviewing the reasons behind incorrect answers is the only way to close gaps

DOK 2: Knowledge Tree

Category 1: Test Format & Scoring Mechanics

- **Source: ETS GRE Subject Test Physics Fact Sheet**
    - **DOK 1 — Facts:**
        - The test is 70 multiple choice questions in 2 hours
        - No penalty for guessing — guessing > not answering
        - Score is 200 to 990; don’t need perfection, usually a 60ish out of 70 gets you a 990
        - May have “diagrams, graphs, experimental data and descriptions of physical situations”
        - “(1) Classical Mechanics, (2) Electromagnetism, and (3) Quantum Mechanics and Atomic Physics” are separate subscores
        - Computer-based administration beginning in September 2023
        - Offered only in April, September, and October; most physics majors take it September or October of their senior year
    - **DOK 2 — Summary:**
        - The three subscores are the “common questions” — learning priority should be heavily weighted toward them
        - Short time crunch makes learning speed a priority
        - Prioritize heuristics over content
    - **Link to source:** https://www.ets.org/pdfs/gre/fact-sheet-physics.pdf

Category 2: Topic & Content Coverage

- **Source: ETS GRE Subject Test Physics Fact Sheet + Practice Book + Conquering the Physics GRE + physicsGRE.com forum**

- **DOK 1 — Facts:**
  - Subjects are Classical Mech, Electromagnetism, Optics/Waves, Thermo+Stat Mech, Quantum Mech, Atom Physics, Special Relativity, Lab Methods, Specialized Topics
  - The bulk of the exam is lower-division (first 3 years of undergrad) material — a former ETS Committee member and forum users both confirm intro calculus-based physics textbooks cover most of what's tested; upper-level topics are a small fraction
  - Some topics have much more emphasis than others
  - High-yield subtopics: in Classical Mechanics — small oscillations and normal modes of coupled oscillators, elastic and inelastic collisions; in Electrodynamics — applications of Gauss's law, boundary conditions for the EM field, dielectrics, polarization; in Quantum Mechanics — selection rules, particle in a box with perturbations, symmetries, harmonic oscillator; in Statistical Physics — partition function, entropy; in Particle Physics — very basic questions
  - There is always at least one question about a Nobel-prize-related topic, such as graphene or 21 cm radiation
  - "single and multivariate calculus, coordinate systems (rectangular, cylindrical and spherical), vector algebra and vector differential operators, Fourier series, partial differential equations, boundary value problems, matrices and determinants, and functions of complex variables" are math concepts common on the exam
- **DOK 2 — Summary:**
  - Generally these are easily accessible via textbooks, however flashcards can help with the instant recall
  - The flashcards should have a "speed mode" where users must recall formulas as quickly as possible (given the AI feature it could go word problem → formula)
  - Interleaving

Category 3: Speed & Time Management

- **Source: ETS Practice Book + Conquering the Physics GRE + APS Video + physicsGRE.com forum + Personal attempt**
  - **DOK 1 — Facts:**
    - Speed is a priority: Spending too much time remembering a formula (physics or math) will be detrimental
    - No calculators
    - Start with easy quick questions to boost your average allowed time
    - Since time is of essence, prioritize what you can answer
    - All questions are of equal value; do not waste time pondering individual questions you find extremely difficult or unfamiliar

    - You may want to work through the test quickly, first answering only the questions about which you feel confident, then going back and answering questions that require more thought, and concluding with the most difficult questions if there is time
    - There are problems in the pGRE that need lengthy calculations and there are trivia questions; trivia questions are either you know them or you don't, so this type should be looked at first, and you should be able to spot questions that take significantly long
    - Problems can be quite wordy, reading alone can be a significant source of time loss
    - When the student tried to answer fast, they fell into obvious traps and made algebra errors; when they tried to be accurate, they ran out of time
  - **DOK 2 — Summary:**
    - Another factor to learn is "question judging" (i.e., a student should be able to identify a question as should skip quickly rather than waste time on it)
    - Supports insight of the app focusing on speed rather than content — several sources already exist to learn physics concepts
    - Common physics sense and heuristics for answering rapidly

Category 4: Problem-Solving Heuristics

- **Source: Conquering the Physics GRE + APS Video + Personal attempt**
  - **DOK 1 — Facts:**
    - Dimensional analysis heuristic (some problems can be easily solved without fully solving, should prioritize this)
    - Limiting cases (Try certain numbers, eliminate what doesn't work)
    - Estimate numerics — especially important as no calc (How can you estimate an equation quickly)
    - A lot of emphasis on the "common sense" practice (quickly eliminate obvious answers)
    - During "variable problems", can substitute numbers for the variables and figure it out from there
    - Multiple problems, however, are just rote memory
  - **DOK 2 — Summary:**
    - These are the skills that should appear in the app, as well as blunt fast memory
    - AI could generate problems quickly, users must answer in quick succession

Category 5: Memorization vs. Derivation Strategy

- **Source: Conquering the Physics GRE + Kirkby + APS Video + physicsGRE.com forum**
  - **DOK 1 — Facts:**

- “Derive, don’t memorize” formulas — too many to memorize; only a select few (hundred) are worthy of memory
- Don’t use other’s formula sheets, make your own
- One student reported that reading textbooks was somewhat a waste of time, because too much time was spent re-deriving the equations
- Memory and Speed > Understanding
- Higher understanding can lead to faster learning
- Knowing the derivation can help to connect concepts mid-exam
- Equations are explained then shown (Generation effect, have students derive it themselves before seeing the equation)

- **DOK 2 — Summary:**
  - Flashcards can be used for instant recall
  - Should have interleaved formulas, to appeal to PGRE format
  - The learning sections should follow a similar style to *Conquering the PGRE* (derive a formula, then explicitly state it)

Category 6: Practice & Study Behavior

- **Source: ETS Practice Book + Kirkby + Jaan Li + physicsGRE.com forum + PrepScholar + APS Video + Edvoy**
  - **DOK 1 — Facts:**
    - Best way to study is to do problem after problem
    - Practice tests are the best way to solve “GRE style” questions
    - Always take tests under testing conditions, exactly as you will on the real thing
    - Tests are early morning, make sure studying happens then to be prepared
    - Take one ETS sample test early in preparation, another before the midway point, and save the last sample exam for a few weeks before the exam to work on speed and time management
    - Successful preparation in one sentence: do the 500 practice problems found online and understand the solutions and how to do them as fast as possible
    - Read the comments of the online solutions, as they frequently give ways of solving problems much faster
    - It is recommended to allow several months of consistent studying, typically ranging from 3 to 6 months
    - After completing practice tests and practice problems, thoroughly review your mistakes and understand the reasons behind the incorrect answers

- The student read “Conquering” start to finish, did all end-of-chapter problems, all practice exams in the book and released 2001–2008 and 2017 tests under real conditions, on a systematic 12-week plan, did Anki flashcards daily, and still plateaued around 800
- Textbooks should be for review/compiling summaries, don’t just learn from scratch
- GRE Subject Test questions are designed to measure skills and knowledge gained over a long period of time. Although you might increase your scores to some extent through preparation a few weeks or months before you take the test, last-minute cramming is unlikely to be of further help
- Random order questions (INTERLEAVING IS HEAVILY NEEDED)
- While studying, stick to a topic (but need to interleave?)
- Concepts build on each other, useful to illustrate “building concept” in this manner
- Physics: A Student Companion — concise overviews of all topics, easy subdivisions for quick topic review, boxed equations for making finding important info easier, not for starting from scratch
- Choosing the right problem to solve is key

- **DOK 2 — Summary:**
    - Problem volume is the core loop; interleaving is the core structure — build within a topic first, then mix across topics as the student advances
    - Practice tests must be taken under real conditions; re-using known tests makes them worthless
    - Reviewing mistakes after every practice session is non-negotiable

Important Caveats for research

- ANY SOURCES BEFORE 2024 IS INACCURATE FORMAT WISE AS THE GRE CHANGED (100 → 70 questions, online format) IN SEPTEMBER 2023. CONTENT MAY STILL BE FINE
- There is no widely-recognized, format-current (post-Sept-2023), Physics-GRE-specific prep book or study guide from a major publisher
- The 3rd edition of Conquering (2018) remains the most-recommended book despite predating the format change
- Most “2025/2026” pages claiming to be updated are recycled SEO content with errors
- The old tests from the 90s are much harder than the current versions
- “Conquering The Physics GRE” tests harder than actual exams
- The single official post-2023 practice test inside the ETS Practice Book is the only released full-length exam matching the current format

    - University physics department pages (e.g. Ohio State) tend to be more reliable than commercial aggregators because departments update them for advising
    - Several sources (Varsity Tutors, Jaan, forum threads) still describe the pre-2023 100-question format — their raw-count claims don’t map directly onto the current test even though the underlying strategy advice still holds
- **Link to source:** https://www.ets.org/pdfs/gre/fact-sheet-physics.pdf